Why Generate a Click? Plug-in Decision Heads for Web Agents

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Paper version: arXiv 2610.20107v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *a choice from a short list: one of eleven action types and one of a few dozen boxes.*

Only typing or answering text is open ended — the rest is a **readout**, not generation.

Motivation: three observations

1 **Written choices.** click(6): one decoding pass per token.

2 **Decision already inside.** Prefill read goal + every box.

3 **Two readouts, one pass.** Heads read; generation keeps new text.

Evaluation setup

WebArena · 174 tasks | MiniWoB++ 119 · 5 seeds | M2W · 3 offline splits | GUIAct · 4 offline sets | Qwen3-VL 4B / 8B + LoRA | 213k steps · 3 seeds

- **Online:** WebArena 174 tasks + MiniWoB++ 119; **offline:** 3 M2W + 4 GUIAct sets.
- WebArena tuning replays 1,298 expert steps on 115 tasks **disjoint from the test tasks**.

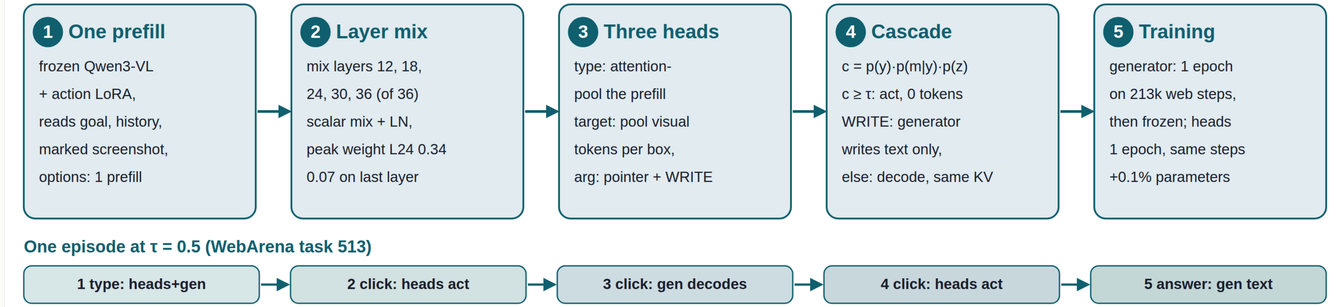
Our read (CUA Radar)

61	★★	online
score of 100	band	evidence level

- **Reports:** online WebArena + MiniWoB++; 3 seeds; same generator + boxes; token + time cost; ablation.
- **Partial:** held-out env partial; judge untested; no code.
- Sub-pattern C05. Authors' numbers, not rerun.

Method: heads on the prefill of a frozen generator

- **Sees / acts:** marked screenshot (numbered boxes), goal, text history, option list; 11 action types, desktop web.
- **Heads (6.0M / 9.2M, ~0.1%):** type pools the prefill; target pools each box's visual tokens; argument points at an option or a WRITE slot.
- **Signal:** heads train 1 epoch on the same 213k steps with the generator frozen; joint training instead costs generation up to 4.9 pts.



Our redraw (not the paper's figure): one prefill feeds both the frozen generator and the plug-in heads.

$$c_t = p(\hat{y} \mid x_t)\; p(\hat{m} \mid \hat{y}, x_t)\; p(\hat{z} \mid \hat{y}, \hat{m}, x_t)$$

action confidence; act at once when $c_t \geq \tau$

Results: online task success (%)

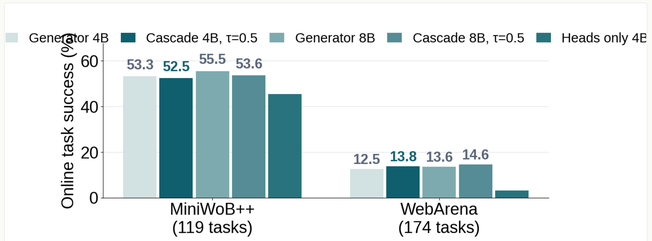


Table 2, redrawn: online task success, mean of 3 seeds, greedy decoding; cascades at τ = 0.5.

WebArena 12.5 → 13.8 (4B), 13.6 → 14.6 (8B), under half the tokens

SO WHAT → *τ = 0.9 keeps MiniWoB++ within 0.2 pts (53.1 / 55.4).*

Ablation: the threshold on M2W unseen websites

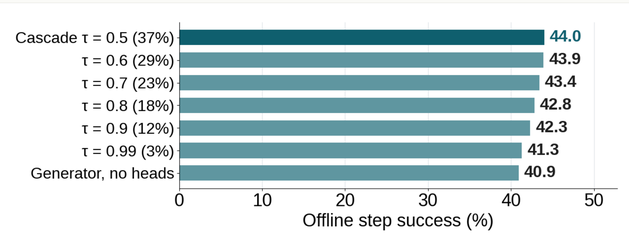


Table 15, redrawn: 4B cascade on M2W cross-website (offline step success); parentheses give the share of steps the heads decide.

SO WHAT → *Looser τ → more steps by the heads → higher success on unseen websites; at τ = 0.5 random routing of the same share would give 39.8%. On GUIAct the paper keeps τ ≥ 0.8; at τ = 0.9 the cascade sits 0.3 below to 1.4 above its generator on all seven offline sets.*

Headline Numbers

13.8%
WebArena success, 4B
vs 12.5% generator

3.1 tok/step vs 7.4

1.39× faster, τ = 0.5

+3.1 pts M2W unseen sites

The paper's own controls

THE PAPER'S OWN CONTROLS	4B	8B
Seen templates (47)	34.8 → 41.8	33.3 → 36.9
New templates (127)	4.2 vs 3.4	6.3 vs 6.3
τ = 0.9, offline sets	0.3 below, 1.4 above	heads decide 12–64%
Joint training cost	−0.5 to −4.9	−0.5 to −4.3

Gain sits in the 47 seen templates

Takeaway

Heads read the choice from the prefill the agent already computes; decoding keeps uncertain steps and new text — under half the tokens.

6.0M / 9.2M head params | +0.1% of backbone

Scan to read the full paper

arxiv.org/abs/2610.20107v1